

Calculators, cell phones, or notes are not allowed.

- (1) Compute the Cyclotomic Polynomials $\varphi_2(x)$, $\varphi_3(x)$, $\varphi_6(x)$ and $\varphi_9(x)$.

$$\varphi_2(x) = \frac{x^2 - 1}{\varphi_1(x)} = \frac{x^2 - 1}{x - 1} = x + 1.$$

$$\varphi_3(x) = \frac{x^3 - 1}{\varphi_1(x)} = \frac{x^3 - 1}{x - 1} = x^2 + x + 1.$$

$$\varphi_6(x) = \frac{x^6 - 1}{\varphi_1(x)\varphi_2(x)\varphi_3(x)} = \frac{x^6 - 1}{(x^3 - 1)\varphi_2(x)} = \frac{x^3 + 1}{\varphi_2(x)} = \frac{x^3 + 1}{x + 1} = x^2 - x + 1$$

$$\varphi_9(x) = \frac{x^9 - 1}{\varphi_1(x)\varphi_3(x)} = \frac{x^9 - 1}{x^3 - 1} = x^6 + x^3 + 1.$$

- (2) Is 6 a square modulo 37 and is 32 a square modulo 41?

Working modulo 37, we have $6^2 = 36 = -1$, $6^4 = 1$, $6^{16} = 1^4 = 1$ and so $6^{(p-1)/2} = 6^{18} = 6^2 \cdot 6^{16} = -1$. Thus 6 is not a square modulo 37.

Similarly working modulo 41, we have $(32)^2 = (-9)^2 = 81 = -1$, $(32)^4 = (-1)^2 = 1$, $(32)^{16} = 1^4 = 1$ and hence $(32)^{(p-1)/2} = 32^{20} = 32^{16} \cdot 32^4 = 1 \cdot 1 = 1$. Thus 32 is a square modulo 41.